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STM32CubeProgrammer 2.23.0

POCKET REFERENCE · GUI & CLI

Practical STM32 programming reference · GUI, SWD and CLI.

01 · GUI versus CLI

i STM32CubeProgrammer is an MCU programming utility. Unlike Eclipse, its documentation does not provide a comparable list of global GUI keyboard shortcuts.

i CLI lines are terminal commands, NOT keyboard shortcuts.

02 · GUI: connection and memory

ST-LINK > SWD	Select SWD; check SWDIO, SWCLK, GND and target power.
Connect / Disconnect	Connect to or disconnect from the MCU.
Memory & File Edition	Inspect memory at an address; open firmware files.
Erasing & Programming	Erase, Program / Download and Verify; confirm file and address.
Option Bytes (OB)	Configure boot and protection options with care.
Automatic Mode	Set up a repeatable programming sequence.

i GUI labels depend on the MCU and connection mode.

03 · CLI: essential commands

STM32_Programmer_CL	Windows CLI executable; use PATH or a

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I	full file path.
-l	List available interfaces or probes.
-c port=SWD	Connect through ST-LINK over SWD.
-w file.bin 0x08000000	Write a binary to the specified address.
-v	Verify written contents (after -w).
-e all	Erase All accessible Flash sectors. CAUTION.
-rst	Reset the connected MCU.
-hardRst	Hardware Reset if RESET is wired.
-g 0x08000000	Go: start execution from this address.
-r32 0x08000000 0x100	Read 32-bit memory over an address range.

i 0x08000000 is a common example; consult the target STM32 Memory Map.

04 · CLI: Windows examples

Connect	STM32_Programmer_CLI.exe -c port=SWD
Program + Verify	STM32_Programmer_CLI.exe -c port=SWD -w firmware.bin 0x08000000 -v
Read Flash	STM32_Programmer_CLI.exe -c port=SWD -r32 0x08000000 0x100
Reset MCU	STM32_Programmer_CLI.exe -c port=SWD -rst
Run code	STM32_Programmer_CLI.exe -c port=SWD -g 0x08000000

i Run in PowerShell or CMD. .hex / .elf files often encode load addresses.

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05 • Quick troubleshooting

Cannot connect	Check power, GND, SWDIO, SWCLK, ST-LINK Firmware and Reset Mode.
Readout protection	RDP may block reads; protection changes may erase data.
Verify failed	Check Flash address, protection, supply and connections.
Wrong address	Check the Memory Map, Linker Script and Boot configuration.
Firmware not running	Review Vector Table, BOOT, Reset, Clock and TrustZone when relevant.

06 • Terms

SWD / JTAG	Debug interfaces.
ST-LINK	ST programming and debug probe.
Flash / SRAM / OTP	Non-volatile memory / RAM / one-time-programmable memory.
OB / RDP / WRP	Option Bytes / Readout Protection / Write Protection.
DFU / UART / USB	Device Firmware Upgrade / serial port / USB.
CLI / GUI	Command-Line Interface / Graphical User Interface.
Erase / Program / Verify	Remove / write / check memory contents.

07 • Safer programming workflow

1	Connect ST-LINK and power the MCU.
2	Confirm Device ID and Memory Map.
3	Select .elf / .hex / .bin and the .bin

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	address.
4	Back up required data before Erase or OB changes.
5	Perform Program + Verify, then Reset / Go.
6	If it fails, inspect the Log and read back memory.

i Caution: RDP Level 2, OTP and some Secure Programming settings are irreversible.

Sources and verification

ST UM2609 — STM32CubeIDE User Guide; ST STM32CubeProgrammer 2.23 documentation.
https://www.st.com/en/development-tools/stm32cubeide.html
https://www.st.com/en/development-tools/stm32cubeprog.html

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